

SCPC 제출 모델

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Task Overview

■ Task

1. A multimodal multiple-choice question answering task

- Select the correct answer from four options based on a given image and textual query



Figure: An image

Textual query:

What might be the purpose of the person's workout in the image?

Four options:

- A. Building muscle and strength
- B. Practicing for a marathon
- C. Training for a cycling race
- D. Preparing for a swimming competition

Answer:

A

Proposed Method

Method

1. Ensemble Multimodal Reasoning for Visual Multiple-Choice QA with CLIP* and SigLIP**

- Pre-trained CLIP: "laion/CLIP-ViT-g-14-laion2B-s12B-b42K" (HuggingFace)
- Pre-trained SigLIP: "google/siglip-so400m-patch14-384"(HuggingFace)
- Total number of parameters from both models: 2.2B

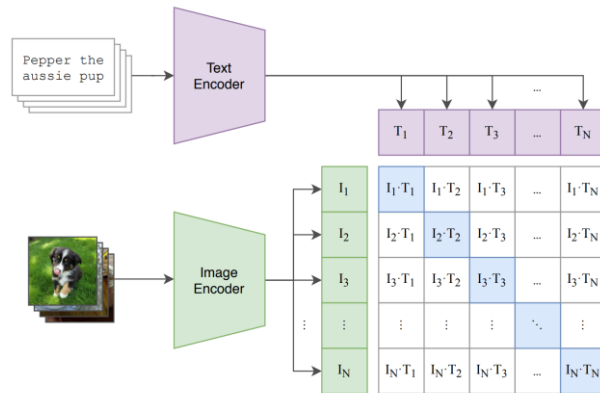


Figure: CLIP

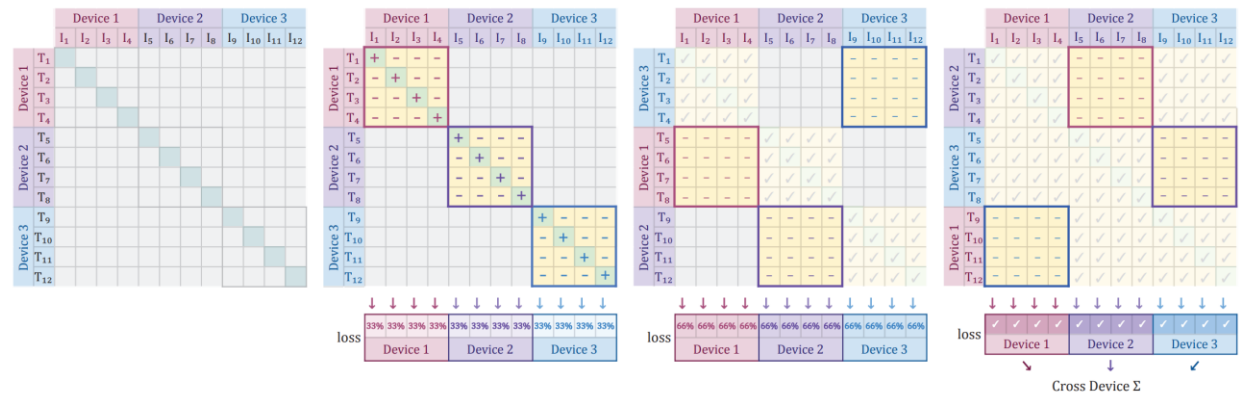


Figure: SigLIP

* Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PmLR, 2021.

** Zhai, Xiaohua, et al. "Sigmoid loss for language image pre-training." Proceedings of the IEEE/CVF international conference on computer vision. 2023.

■ Method

1. Zero-shot multi-class classification

- CLIP: Dual encoder-based multimodal model
 - Image encoder: Embedding image: $\mathbf{i} \in \mathbb{R}^d$
 - Text encoder: Embedding text: $\mathbf{t}_c \in \mathbb{R}^d$, for $c \in \{A, B, C, D\}$
 - Cosine similarity: $s_c = \langle \mathbf{i}, \mathbf{t}_c \rangle$
- SigLIP: Operates in the same way as CLIP

2. Scoring

- CLIP
 - $\text{argsort}(s) = [c_1, c_2, c_3, c_4]$ where $(s_{c_1} \geq s_{c_2} \geq s_{c_3} \geq s_{c_4})$
 - $\text{score}_{c_1} = 4, \text{score}_{c_2} = 3, \text{score}_{c_3} = 2, \text{score}_{c_4} = 1$
 - $\text{score}_{dict} = \{A: \text{score}_A, B: \text{score}_B, C: \text{score}_C, D: \text{score}_D\}$
- SigLIP: Operates in the same way as CLIP

▪ Method

1. Zero-shot multi-class classification

- Ensemble
 - Final score_c = clip score_c + siglip score_c, $\forall c \in \{A, B, C, D\}$
 - Prediction = $\{c \mid \text{final score}_c = \max_k \text{final score}_k\}$
 - If multiple candidates, Choose the one with the highest CLIP score